

# PAPER\_279 Sombrero SMBH Dominance Ratio $\hat{r}_{\text{BH}}^3$ and UQFF Sphere of Influence $r_{\text{SOI}}$

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**Session:** 77 â€” March 2026 **Framework:** Unified Quantum Field Framework (UQFF 2.0) **Status:** Complete â€” embedded in SOMBRERO\_UQFF\_MODULE.cpp  
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## Abstract

The Sombrero Galaxy (M104) harbours one of the most massive black holes relative to its host galaxy mass of any object in the Local Universe:  $M_{\text{BH}} = 10^{-4} M_{\text{sun}}$  in a galaxy of total mass  $M = 10^8 M_{\text{sun}}$ , giving a **SMBH Dominance Ratio**  $\hat{f}_{\text{BH}} = M_{\text{BH}}/M = 0.01$  (1%). For comparison, the Milky Way's Sgr A\* has  $\hat{f}_{\text{BH}} \approx 4 \times 10^{-8}$  ( $\sim 0.004\%$ ); Sombrero's SMBH is **250 $\times$  more dominant relative to its host**. Within the UQFF framework, we define the **UQFF Sphere of Influence**  $r_{\text{SOI}} = r_{\text{A}}(\hat{f}_{\text{BH}})$ , the radius at which the central BH's direct gravitational contribution equals the total galaxy contribution at the reference radius. For Sombrero,  $r_{\text{SOI}} = 2.36 \times 10^4 \text{ m}$   $\approx$  a precise UQFF prediction setting the boundary inside which BH gravity exceeds galaxy-mean gravity in the UQFF model.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4$  day $^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. Observational Motivation

## 1.1 The Sombrero SMBH

The Sombrero Galaxy's central black hole mass has been measured through stellar and gas kinematics:

- Ford et al. (1996):  $M_{\text{BH}} = (1.0 \pm 0.75) \times 10^6 M_{\text{sun}}$  (gas kinematics, HST)
- Kormendy et al. (1996): confirmation from stellar dynamics
- Jardel et al. (2011):  $M_{\text{BH}} \approx 6.6 \times 10^6 M_{\text{sun}}$  (JAM modelling, consistent with  $\sim 10^6 M_{\text{sun}}$  range)

We adopt  $M_{\text{BH}} = 1.0 \times 10^1 M_{\text{sun}} = 1.989 \times 10^3 \text{ kg}$ .

The total galaxy mass (stars + gas + DM within the reference radius) is  $M = 10^{11} M_{\text{sun}} = 1.989 \times 10^{47} \text{ kg}$ .

## 1.2 Why $\hat{\mathbb{I}}^3$ BH Matters

In standard astrophysical models, BH sphere-of-influence calculations use the velocity dispersion ( $r_{\text{SOI}} = GM_{\text{BH}}/\hat{\sigma}^2$ ). The UQFF framework generalises this to a mass-ratio-based definition consistent with the 26-layer Triadic gravity structure, yielding

a dimensionless parameter  $\hat{I}^3_{\text{BH}}$  that naturally encodes the BH's dominance within the UQFF vacuum field hierarchy.

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## 2. Theoretical Derivation

### 2.1 SMBH Dominance Ratio

We define the dimensionless **SMBH Dominance Ratio**:

$$\gamma_{\text{BH}} = \frac{M_{\text{BH}}}{M}$$

For the Sombrero Galaxy:

$$\gamma_{\text{BH}}^{\text{Sombrero}} = \frac{10^9 M_{\odot}}{10^{11} M_{\odot}} = 0.01 = 1\%$$

### 2.2 SMBH Direct Gravitational Contribution

The BH contribution to  $g_{\text{total}}$  at the reference radius  $r$ :

$$g_{\text{BH}} = \frac{GM_{\text{BH}}}{r^2} = \gamma_{\text{BH}} \cdot \frac{GM}{r^2} = \gamma_{\text{BH}} \cdot g_{\text{base}}$$

Substituting  $\hat{I}^3_{\text{BH}} = 0.01$  and  $g_{\text{base}} = 2.382 \times 10^{-10} \text{ m/s}^2$ :

$$g_{\text{BH}} = 0.01 \times 2.382 \times 10^{-10} = 2.382 \times 10^{-12} \text{ m/s}^2$$

### 2.3 UQFF Sphere of Influence

The **UQFF Sphere of Influence**  $r_{\text{SOI}}$  is defined as the radius  $r' < r$  where the BH gravitational contribution equals the total galaxy contribution at  $r$ :

$$g_{\text{BH}}(r') = g_{\text{base}}(r)$$

$$\frac{GM_{\text{BH}}}{r'^2} = \frac{GM}{r^2}$$

Solving:

$$r'^2 = r^2 \cdot \frac{M_{\text{BH}}}{M} = r^2 \cdot \gamma_{\text{BH}}$$

$$r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$$

For Sombrero:

$$r_{\text{SOI}} = 2.36 \times 10^{20} \cdot \sqrt{0.01} = 2.36 \times 10^{20} \times 0.1 = 2.36 \times 10^{19} \text{ m}$$

**Physical interpretation:** Within  $r_{\text{SOI}} = 2.36 \times 10^{19} \text{ m}$  ( $\sim 2.49 \text{ kly}$ ), the direct BH gravitational acceleration exceeds the galaxy-mean  $g_{\text{base}}$ . This is the UQFF-predicted boundary of BH gravitational dominance.

## 2.4 Verification

At  $r' = r_{\text{SOI}}$ :

$$g_{\text{BH}}(r_{\text{SOI}}) = \frac{GM_{\text{BH}}}{r_{\text{SOI}}^2} = \frac{G \cdot 0.01M}{(0.1r)^2} = \frac{0.01 \cdot GM}{0.01 \cdot r^2} = \frac{GM}{r^2} = g_{\text{base}}$$

## 3. Comparative SMBH Dominance Table

| Galaxy                 | $M_{\text{BH}} (M_{\text{sun}})$ | $M_{\text{total}} (M_{\text{sun}})$ | $\hat{I}^3_{\text{BH}}$   | $r_{\text{SOI}} / r_{\text{ref}}$ |
|------------------------|----------------------------------|-------------------------------------|---------------------------|-----------------------------------|
| Milky Way (Sgr A*)     | $\sim 4 \times 10^6$             | $\sim 10^{11}$                      | $\sim 4 \times 10^{-10}$  | $\sim 6.3 \times 10^3$            |
| Andromeda (M31)        | $\sim 1.4 \times 10^7$           | $\sim 10^{12}$                      | $\sim 1.4 \times 10^{-9}$ | $\sim 1.2 \times 10^4$            |
| M87                    | $\sim 6.5 \times 10^9$           | $\sim 6 \times 10^{11}$             | $\sim 1.1 \times 10^{-8}$ | $\sim 3.3 \times 10^4$            |
| <b>Sombrero (M104)</b> | <b><math>\sim 10^9</math></b>    | <b><math>\sim 10^{11}</math></b>    | <b>0.01</b>               | <b>0.1</b>                        |

**Sombrero's  $\hat{I}^3_{\text{BH}} = 0.01$  is the highest of any nearby well-measured galaxy in the UQFF catalogue, making it the dominant test-case for SMBH-galaxy UQFF coupling.**

Key comparison ratios: - Sombrero/Sgr A\*:  $\hat{I}^3_{\text{BH}}$  ratio =  $0.01 / 4 \times 10^{-10} = 250$   
- Sombrero/M87:  $\hat{I}^3_{\text{BH}}$  ratio =  $0.01 / 1.1 \times 10^{-8} = 9$   
- Sombrero/Andromeda:  $\hat{I}^3_{\text{BH}}$  ratio =  $0.01 / 1.4 \times 10^{-9} = 71$

## 4. UQFF BH Contribution in the Master Equation

The BH term enters `computeG()` as an additive contribution alongside the 26-layer Triadic sum:

$$g_{\text{total}} = [\dots + g_{\text{BH}} + \dots] \cdot \kappa_{\text{recession}} \cdot \sigma_{\text{SC}}$$

$$= [\dots + 2.382 \times 10^{-12} \text{ m/s}^2 + \dots] \times 0.99374 \times (1 - 10^{-20})$$

**BH fractional contribution to  $g_{\text{total}}$**  (estimated):

$$\frac{g_{\text{BH}}}{g_{\text{total}}} \approx \frac{2.382 \times 10^{-12}}{1.238 \times 10^{-8} + 2.382 \times 10^{-12} + \dots} \approx \frac{2.382 \times 10^{-12}}{1.24 \times 10^{-8}} \approx 1.9 \times 10^{-4}$$

While the BH's direct gravitational contribution at the reference radius  $r$  is a small fraction of the 26-layer Triadic sum ( $\sim 0.019\%$ ), the  $r_{\text{SOI}}$  formula reveals that inside  $2.36 \times 10^3 \text{ m}$ , BH gravity dominates the reference-point baseline – a qualitative distinction for UQFF predictions of inner galactic structure.

## 5. Module Implementation

```
// PAPER_279 – SOMBRERO_UQFF_MODULE.cpp, updateCache()
gamma_BH = M_BH / M; // 0.01 = 1%
r_SOI     = r * std::sqrt(gamma_BH); // 2.36e20 Å = 0.1 = 2.36e19 m

// Applied in computeG():
double g_BH = G_grav * M_BH / (r * r); // = gamma_BH * g_base = 2.382e-12 m/s^2
```

**Computed values for Sombrero M104:**

| Quantity                                                | Value                                 | Units          |
|---------------------------------------------------------|---------------------------------------|----------------|
| $M_{\text{BH}} = 10^3 M_{\text{sun}}$                   | $1.989 \times 10^3 M_{\text{sun}}$    | kg             |
| $M = 10^4 M_{\text{sun}}$                               | $1.989 \times 10^4 M_{\text{sun}}$    | kg             |
| $\hat{I}^3_{\text{BH}} = M_{\text{BH}}/M$               | 0.01                                  | dimensionless  |
| $g_{\text{BH}} = G \cdot M_{\text{BH}}/r^2$             | $2.382 \times 10^{-12} \text{ m/s}^2$ | $\text{m/s}^2$ |
| $r_{\text{SOI}} = r \cdot \sqrt{\hat{I}^3_{\text{BH}}}$ | $2.36 \times 10^3 \text{ m}$          | m              |
| $r_{\text{SOI}}$ in kly                                 | $\sim 2.49$                           | kly            |

## 6. Key Constants Introduced

| Symbol                  | Value                                 | Units          | Description                                    |
|-------------------------|---------------------------------------|----------------|------------------------------------------------|
| $\hat{I}^3_{\text{BH}}$ | 0.01                                  | dimensionless  | SMBH Dominance Ratio<br>$M_{\text{BH}}/M$      |
| $r_{\text{SOI}}$        | $2.36 \times 10^3 \text{ m}$          | m              | UQFF Sphere of Influence<br>radius             |
| $g_{\text{BH}}$         | $2.382 \times 10^{-12} \text{ m/s}^2$ | $\text{m/s}^2$ | BH direct gravitational<br>contribution at $r$ |

## 7. Physical Significance

1. **SMBH Dominance Ratio as a universal UQFF parameter:**  $\hat{I}^3_{BH} = M_{BH}/M$  provides a single dimensionless number characterising how BH-dominated a galaxy is. It ranges from  $\sim 10^{-8}$  (Sgr A\*) to  $\sim 0.01$  (Sombrero), spanning three orders of magnitude in the current UQFF catalogue.
  2. **UQFF Sphere of Influence formula:**  $r_{SOI} = r_{\hat{I}^3_{BH}}$  is derived directly from setting  $g_{BH}(r') = g_{base}(r)$ . This provides a clean, parameter-free prediction for the BH dominance scale in any UQFF module with a known  $M_{BH}/M$  ratio.
  3. **Sombrero as extreme BH-galaxy system:** With  $\hat{I}^3_{BH}$  250% larger than the Milky Way's, Sombrero tests UQFF behaviour in the BH-dominated regime. The large  $\hat{I}^3_{BH}$  implies that inner BH effects begin influencing the 26-layer Triadic structure at radii as large as  $r_{SOI} = 2.36 \times 10^1$  m.
  4. **AGN feedback implications:** The large  $M_{BH}$  implies strong AGN feedback potential. UQFF predicts that AGN activity modifies the vacuum energy density locally, which would appear in the UQFF framework as a time-varying component of  $U_{g1\_vec}[i]$  for the innermost layers a direction for future UQFF module development.
  5. **Generalisation:** The formula  $\hat{I}^3_{BH} = M_{BH}/M$  and  $r_{SOI} = r_{\hat{I}^3_{BH}}$  define a universal UQFF BH dominance prescription applicable to any galaxy module. Future UQFF modules for BH-dominated systems (e.g., NGC 1277, M87) should include this pair of parameters as standard.
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## 8. References

- Ford, H. C. et al. (1996). ApJ, 458, 132. (Sombrero  $M_{BH}$  from gas kinematics, HST)
  - Kormendy, J. et al. (1996). ApJ, 459, L57. (Sombrero BH mass confirmation)
  - Jarrel, J. R. et al. (2011). ApJ, 739, 21. (Sombrero DM halo and BH mass)
  - Gálvez, K. et al. (2009). ApJ, 698, 198. ( $M_{BH}$  correlation)
  - PAPER\_277: UQFF Gravitational Recession Damping  $\hat{I}^0_{recession}$  for  $z = +0.0063$
  - PAPER\_278: Sombrero Dust Ring UQFF Gravitational Ring Resonator  $\hat{I}^0_{ring}$
  - SOMBRERO\_UQFF\_MODULE.cpp (UQFF 2.0, Session 77)
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*UQFF 2.0 a The SMBH Dominance Ratio  $\hat{I}^3_{BH} = M_{BH}/M$  and UQFF Sphere of Influence  $r_{SOI} = r_{\hat{I}^3_{BH}}$  are new universal parameters for UQFF galaxy modules, first derived and tested on the Sombrero Galaxy. a Daniel T. Murphy, Session 77, March 2026.*